

Computer Graphics: Lecture 1

Welcome to Real-time 3D!

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Welcome!

You have done some cool things as a computer science major:

- You have learned and implemented a wide range of algorithms...
- Data structures too...
- You have developed user-facing applications (GUIs)

You have learned many languages, and built many projects, to do all sorts of cool things.

However, there is a set of capabilities inside your computer that you have not even *begun* to unlock. The computer is **way** more powerful than you've been able to take advantage of.

The computer's superpower

Almost every modern computer (including phones) contains a powerful **co-processor** called a GPU—a graphics processing unit. This GPU might be on a special expansion card, like this RTX 5090 here:



The GPU is incredibly powerful, often an order of magnitude more powerful than the CPU.



Questions?